



MACHINE LEARNING

MKT3434

Betül DAŞDEMİR

20067502

Table of Contents

Introduction	2
Developments Made and General Features of the User Interface.....	2
Mathematical Application	3
Results and Screenshots	4
Conclusion.....	9

Introduction

The aim of this project is to create a machine learning platform that integrates dimensionality reduction and clustering algorithms into a user-friendly GUI, making them accessible for real-world applications such as robotics, mechatronics, and image processing. Through this interface, users can select methods like PCA, LDA, t-SNE, and UMAP, allowing for data preprocessing, modeling, and visualization in an interactive and practical environment.

Developments Made and General Features of the User Interface

Key Features and Implemented Modules

- Modern Interface Developed with PyQt6
- Options for data loading and splitting (train/test split, k-fold cross-validation)
- Dimensionality Reduction Tab: PCA, LDA, t-SNE, UMAP
- Clustering Tab: K-Means algorithm with Elbow Method visualization and clustering quality evaluation
- Model Validation: K-Fold Cross-Validation with Accuracy, MSE, and RMSE reporting
- Visualization: Explained Variance Plot, Interactive 2D/3D Scatter Plots (via Plotly)

New Tab: Dimensionality Reduction

- PCA (Principal Component Analysis)
Users can select the number of components; explained variance ratio is visualized via plots. Both 2D and 3D projections are supported.
- LDA (Linear Discriminant Analysis)
Supervised dimensionality reduction is performed using class labels. Visual support is provided with class separation metrics (e.g., Fisher Score).

- t-SNE & UMAP

Hyperparameters such as perplexity and learning rate can be adjusted. Results are visualized in interactive 2D or 3D plots using Plotly.

Clustering

- K-Means

Users can specify the value of k (number of clusters). The optimal k is suggested via the Elbow Method. Clustering quality is evaluated using metrics like Silhouette Score.

- Silhouette Score

Clustering performance is quantitatively evaluated and interpreted.

Data Splitting & Cross-Validation

- Training/Validation/Test ratios (e.g., 70-15-15) are user-adjustable
- K-Fold Cross-Validation is supported (e.g., $k=5$)
- For k -fold validation, average and standard deviation values of Accuracy, MSE, and RMSE are reported

Libraries Used

- PyQt6
- Scikit-learn
- NumPy, Pandas
- Matplotlib, Plotly
- UMAP-learn

Mathematical Application

Given the covariance matrix:

$$\Sigma = \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$$

The eigenvalues and corresponding eigenvectors of this matrix are computed, and the eigenvector associated with the largest eigenvalue is selected. The data is then projected onto this eigenvector to reduce its dimensionality to 1D.

Eigenvalue and Eigenvector Calculation:

- Eigenvalues: $\lambda_1 = 6$, $\lambda_2 = 2$
- Eigenvectors:
 - For λ_1 : $v_1 \approx [0.8507, 0.5257]$
 - For λ_2 : $v_2 \approx [-0.5257, 0.8507]$

Projection:

The dataset can be reduced to either 1D or 2D by projecting it onto these eigenvectors, depending on the desired level of dimensionality.

Results and Screenshots

Below are the screenshots taken from the code output along with their corresponding descriptions.

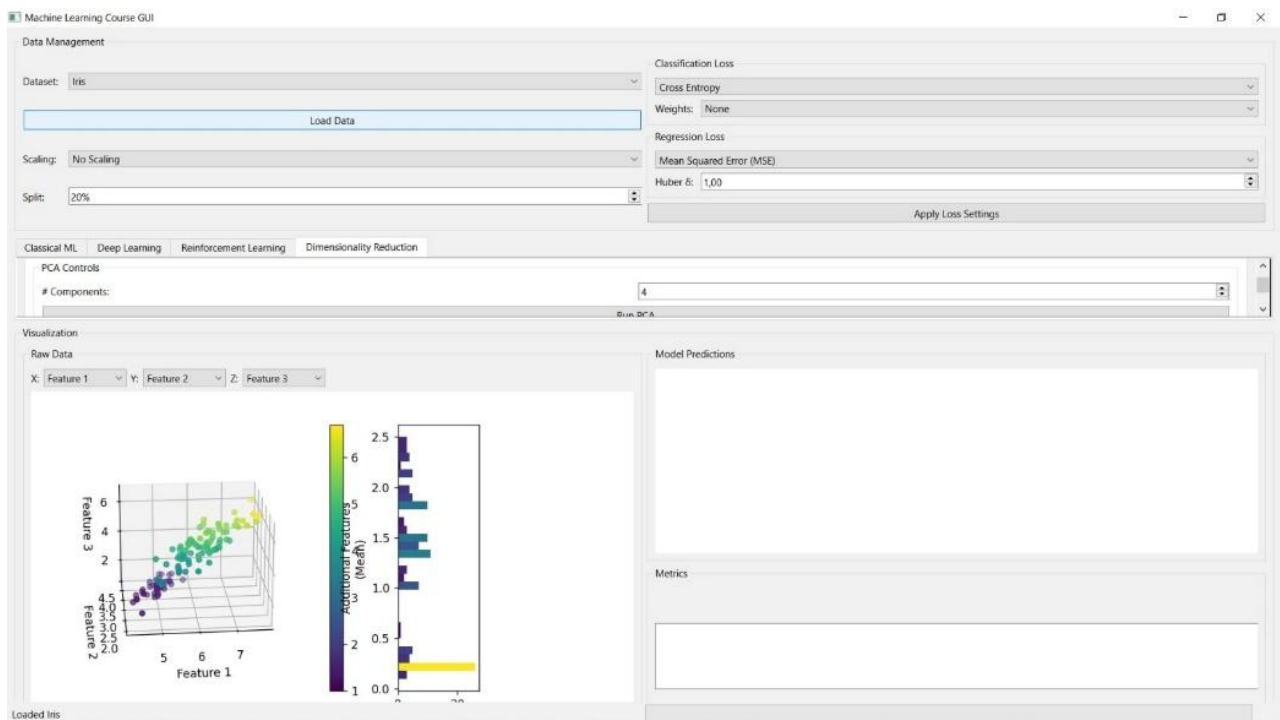


Figure 1

Developed PyQt6-Based Machine Learning Interface

In this interface, operations such as dataset loading, scaling method selection, train-test split ratio adjustment, and loss function configuration are presented in a user-friendly manner. The top menu allows switching between different learning modules

such as "Classical ML", "Deep Learning", "Reinforcement Learning", and "Dimensionality Reduction". While the bottom-left section displays 3D data visualizations and histograms, the right panel presents model outputs and performance metrics. This integrated structure enables users to carry out the modeling and analysis process in a visually rich and interactive environment.

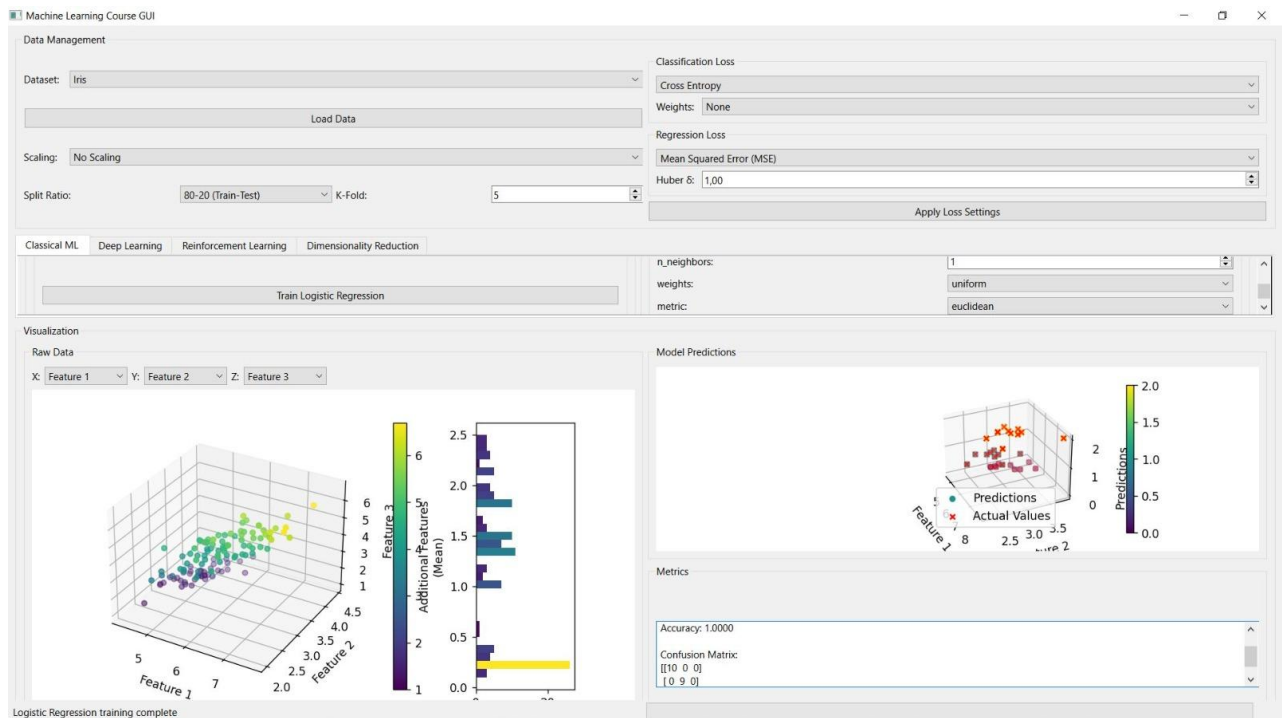


Figure 2

Machine Learning Model Training and Prediction Visualization

In this screen, the model has been successfully trained using the Iris dataset, and its predictions on the test data have been visualized in a 3D plot. The bottom-right panel displays the model's accuracy and the confusion matrix. In this particular example, the model achieved an accuracy of 1.000 (100%). Additionally, the bottom-left panel presents a visualization of the raw feature data. This setup allows users to simultaneously analyze both the model outputs and the associated performance metrics in real time.

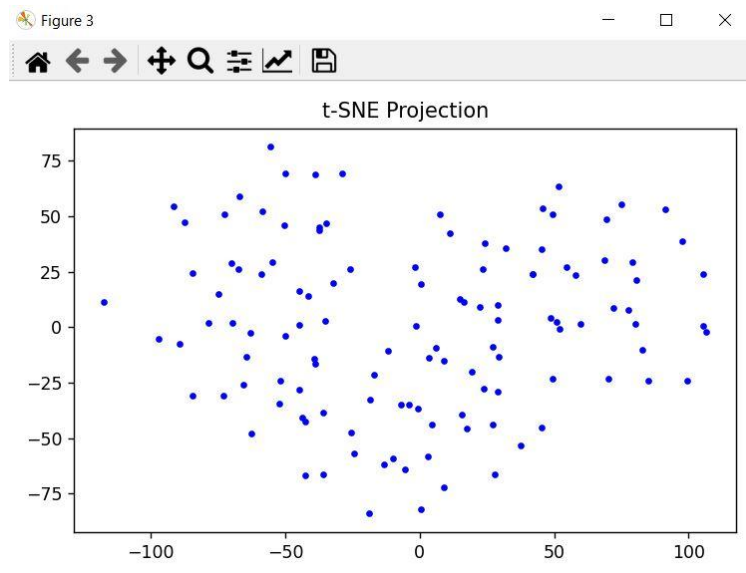


Figure 3

2D Data Projection with t-SNE

In the graph, high-dimensional data has been reduced to two dimensions using the t-SNE algorithm. t-SNE provides intuitively meaningful visualizations by grouping similar instances together, especially in datasets with complex structures. However, class labels are not shown in this particular visualization. When color-coded class distinctions are added, the algorithm's ability to separate inter-class structures becomes more apparent. While t-SNE is powerful for visual analysis, it has limitations in terms of interpretability.

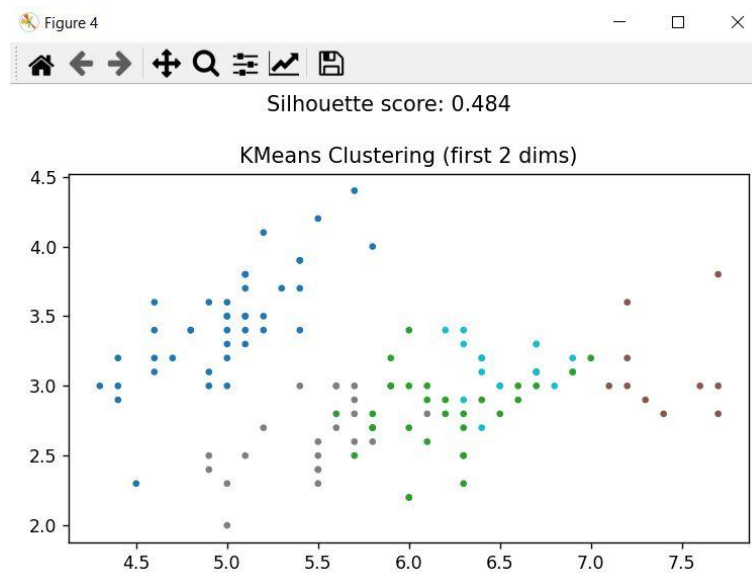
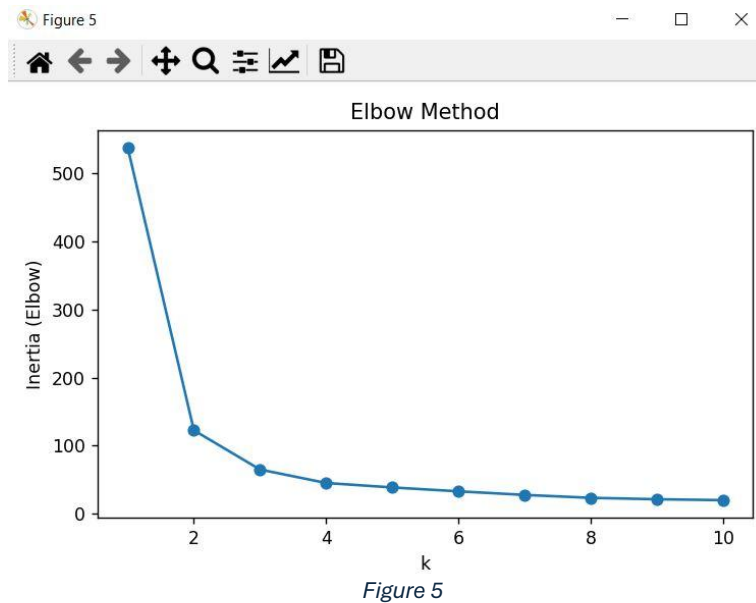


Figure 4

K-Means Clustering Result (Visualization in First 2 Dimensions)

In the graph, clusters identified by the K-Means algorithm are displayed in color. This projection, based solely on the first two features, allows for intuitive observation of the separation between clusters. The Silhouette score calculated to evaluate clustering

performance is 0.484, indicating a reasonable level of separation and internal consistency among the clusters. A Silhouette score close to 0.5 is generally considered to reflect a meaningful clustering structure.



Determining the Optimal Number of Clusters Using the Elbow Method

The graph shows the total within-cluster sum of squares (inertia) calculated for different values of k . The curve forms a noticeable "elbow" around $k = 3$, indicating that this is the optimal number of clusters. The Elbow Method helps prevent over-segmentation in the K-Means algorithm and guides the selection of the most appropriate cluster count for the dataset.

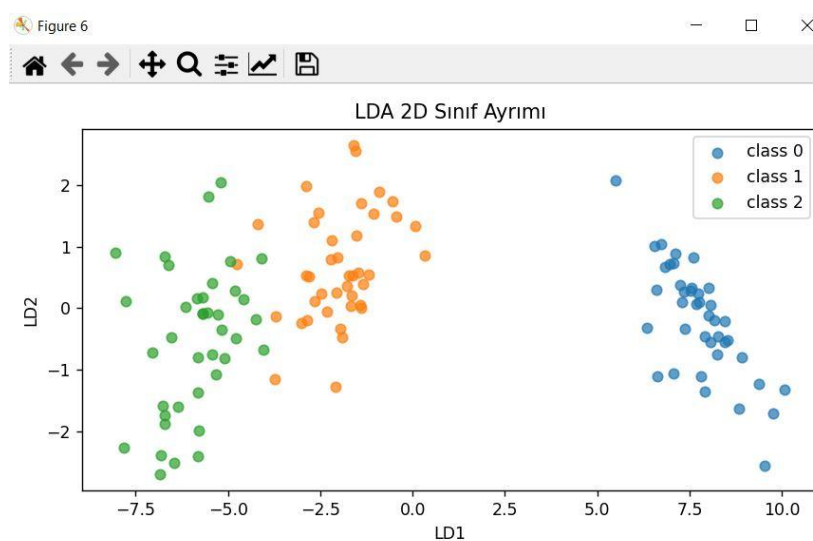


Figure 6

2D Class Separation Visualization with LDA

In the graph, the data has been projected onto two discriminant components (LD1 and LD2) using the LDA method. Each point represents a sample, and the colors indicate different classes (class 0, class 1, class 2). LDA creates this projection by maximizing the separation between classes. As seen in the visualization, the three classes are clearly distinguished from one another. This demonstrates that LDA is an effective method for supervised dimensionality reduction.

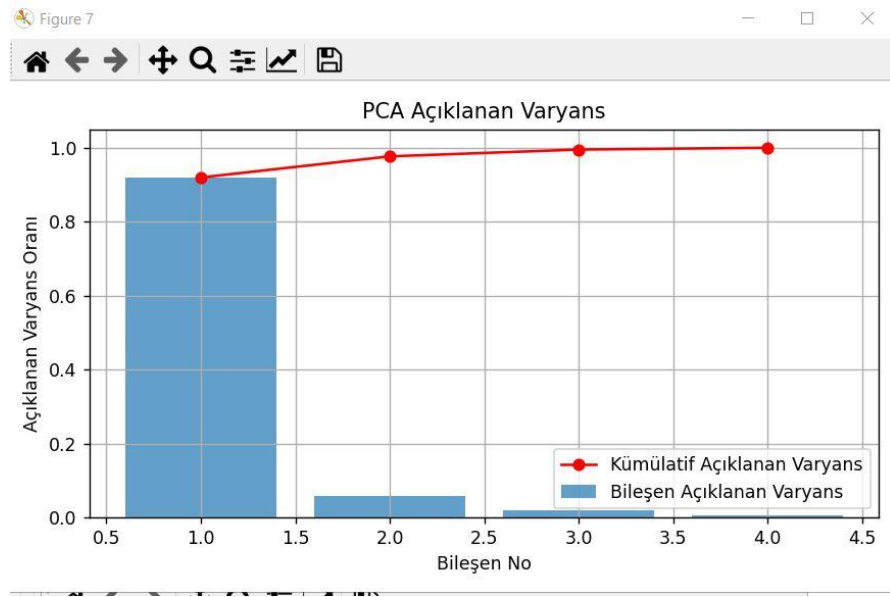


Figure 7

Visualization of Explained Variance Ratios with PCA

In the graph, the blue bars represent the variance explained by each individual principal component, while the red line shows the cumulative explained variance. It can be observed that the first component alone explains approximately 90% of the total variance. The first two components together account for nearly all of the variance in the dataset. This indicates that using only one or two components for dimensionality reduction would be sufficient, with minimal loss of information.

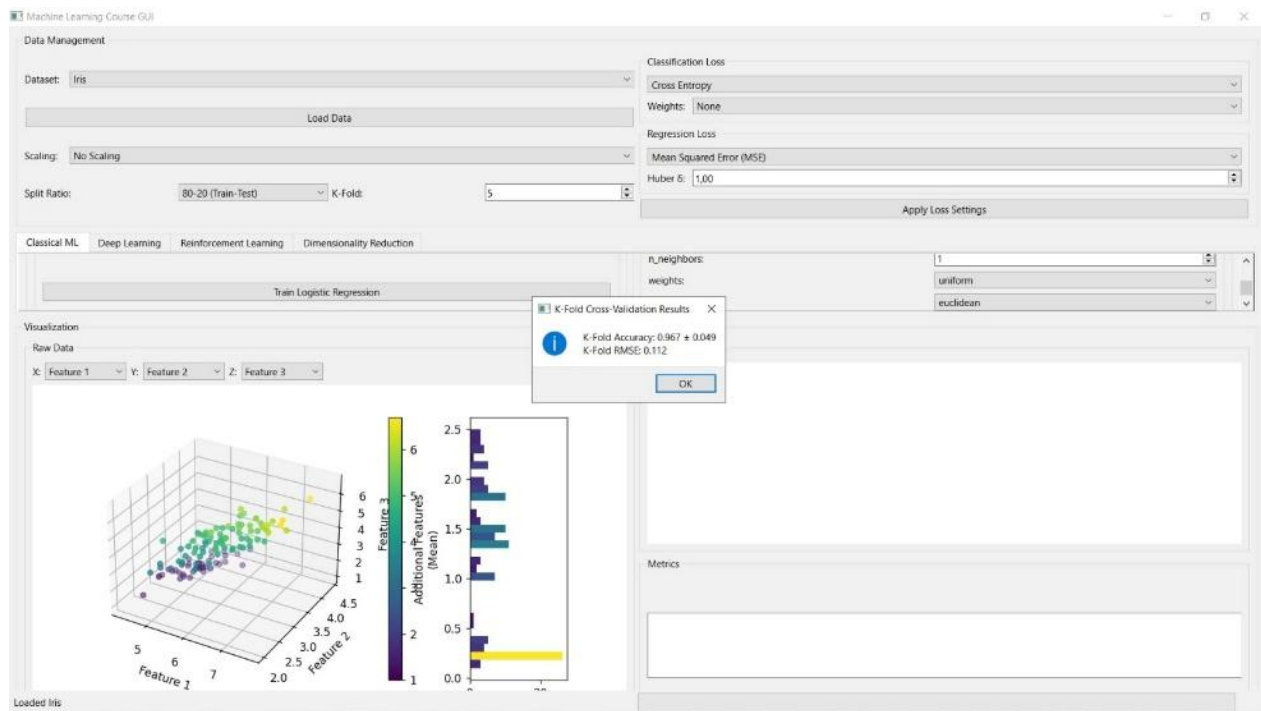


Figure 8

Display of K-Fold Cross-Validation Results

In this screen, the results of 5-fold ($k=5$) cross-validation applied to the Iris dataset are presented to the user via a pop-up window. The calculated average accuracy is 0.849, and the Root Mean Squared Error (RMSE) is 0.112. This structure allows the user to assess the generalization performance of the model and observe potential risks of overfitting or underfitting. The entire process is carried out automatically through the GUI, without requiring manual user intervention.

Conclusion

The developed interface supports both supervised and unsupervised dimensionality reduction techniques, enabling users to meaningfully analyze high-dimensional datasets. Additionally, it provides an integrated analytical environment with model evaluation metrics, data splitting options, and clustering techniques.